

Collaboratively Mitigating Racial Disparities in Automated Speech Recognition and Language Technologies with African American English Speakers

Community-Collaborative and Equity-Centered Approaches Toward Designing Inclusive Natural Language Systems

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ABSTRACT

Automated speech recognition (ASR) systems that rely on natural language processing (NLP) techniques are becoming increasingly prevalent within people's everyday lives. From virtual assistants integrated into mobile devices (e.g. Apple's Siri), smart home assistants (e.g. Google's Nest; Amazon's Alexa/Echo), and vehicles (e.g. Apple's CarPlay; Android Auto); to software tasks such as automatic translation, automatic captioning, automatic subtitling and even hands-free computing, ASR systems are core components of IoT (Internet of things) devices and applications. However, scholars have begun to show that with these increasing innovations and system capabilities, emerges fairness-related harms of user experiences and racial disparities that negatively impact African American speakers of African American Vernacular English (AAVE). As users of ASR, AAVE speakers' language is less accurately recognized and processed, leading to inequitable interactions among this ethnolect community. My graduate research seeks to address this challenge by developing and validating community-collaborative methods to innovate responsible approaches toward designing more representative and equitable ASR language technologies that accommodate African American speakers of AAVE.

CCS CONCEPTS

• **Human-centered computing** → Collaborative and social computing; Empirical studies in collaborative and social computing;

KEYWORDS

Human Computer Interaction (HCI), Responsible Artificial Intelligence (AI), Natural Language Processing, Inclusive Design, Participatory Design, Co-Design, Community-Based Participatory Research, Action Research, Equity Centered Design

ACM Reference Format:

Jay L. Cunningham. 2023. Collaboratively Mitigating Racial Disparities in Automated Speech Recognition and Language Technologies with African American English Speakers: Community-Collaborative and Equity-Centered

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3577057>

Approaches Toward Designing Inclusive Natural Language Systems. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3544549.3577057>

1 DOCTORAL RESEARCH OVERVIEW AND GOALS

I am a fourth-year doctoral candidate in the Department of Human Centered Design & Engineering (HCDE) at the University of Washington, advised by Professors Dr. Julie Kientz and Dr. Daniela Rosner. With a background in software engineering, product management, and UX research, I am passionate about responsibility/fairness in AI/ML systems and user experiences through inclusive design and algorithmic/design justice. My doctoral research investigates the integration of human-centered design methods with community-collaboration and equity-centered approaches to explore and address the sociotechnical implications of race, culture, identity & power and the intersection of intelligent systems and machines. My graduate work aims to advance and sustain responsible design practices that can lead to more inclusive and equitable user experiences in automated speech recognition and language technologies, with a critical lens on African American English-speaking communities.

My doctoral research and education is supported by the National Science Foundation Graduate Research Fellowship (NSF-GRFP), the Gates Millennium Scholars Program (GMSP-UNCF), the National GEM Consortium Ph.D. Engineering and Science Fellowship (GEM), and the Google Award for Inclusion Research (AIR).

2 BACKGROUND AND MOTIVATION

The HCI research and design communities have long been concerned with the impacts of technology on marginalized groups and communities [19]. From the development of intelligent facial recognition systems [5], prison sentencing technology, hiring algorithms, and healthcare algorithms, artificial intelligence (AI) heavily relies on machine learning (ML) to make critical decisions on the ways in which people can live [35]. These systems are becoming increasingly ubiquitous in the lives of us all and in doing so, we must be prepared to grapple with the intended and unintended implications of them [28]. Even with the altruistic promise of AI to help detect and reduce human bias and prejudice, there exists consequences of AI's "blind-spots" as outlined by scholars Kate Crawford and Ryan Calo [9]. These blind-spots, while broad in categorization, can be seemingly summed as the overlooked, under-evaluated, and

even chauvinistic tendencies of technology researchers, designers, and practitioners to study, innovate, and build technical artifacts without understanding the downsides of interactive systems and intelligent machines [9]. Such systems often reinforce and overwhelmingly benefit the position of those in power who tend to be white, educated, and male [9]. And at the promise of so-called “innovation,” conventional technology design practices themselves “may perpetuate forms of institutional racism and bias,” enabling and legitimizing racialized forms of inequity, as indicated by O’Leary, et al. [38].

Structural issues like anti-blackness, violence against women, and queer-phobia all contribute to patterns of social exclusion and are all being undertaken through development of AI; even when the intent is to mediate or interrupt their impacts, there exists emerging works that focus on the absurdity of solving such issues with technical devices [6, 20]. To date, we’ve increasingly seen the growing contributions among HCI and design scholars highlighting the ways in which these technologies can negatively impact gender minorities, people of color, and other under-represented groups [12, 14, 32]. Scholars like Ruha Benjamin [1], Safiya Noble [24], Ihudiya Ogbonnaya-Ogburu et al. [25], Alex Hanna [16] and so many others characterize a technology design space perpetuating such harms by proposing over-reaching, under-performing, and paradoxically problem-making solutions [10]. Complementing this range of work, interdisciplinary design scholars have brought an intersectional lens to questions of equity [15, 27], citing design as a resource for alternative forms of world-building [4, 36]

The issues of fairness and accountability have been further echoed by corporate ethics boards and civic leaders, with the growing calls for ethics in AI through both practice and legislation. And in further response, technology, and data equity organizations like AI Now, the Algorithmic Justice League, Partnership on AI, and the Design Justice Network are collectively working to highlight and address such disparities through civic engagement and research. Yet despite this rich body of critical analysis, the HCI field continues to struggle with inclusive approaches that effectively address the pervasive and lasting inequitable consequences of racism, bias, sexism, and discrimination perpetuated through AI. And failure to effectively address this invites the potential to wreak havoc on the lives of minority communities and vulnerable groups as they interact with applications and software that we use daily [22].

As asserted by critical technology scholar Arnold Pacey [26] and reiterated by Crawford and Calo [9]:

“Artificial intelligence presents a cultural shift as much as a technical one. This is similar to technological inflection points of the past, such as the introduction of the printing press or the railways. Autonomous systems are changing workplaces, streets and schools. We need to ensure that those changes are beneficial, before they are built further into the infrastructure of everyday life” [26].

And with the rapid emergence of the Internet of Things (IoT), smart devices, and voice-controlled virtual assistants like Amazon’s Echo Alexa, Google Home, and Apple’s Siri as well as various other media and data consumption devices, addressing algorithmic bias, accessibility, access, and inclusion has never been more urgent than now [18, 37].

3 RESEARCH OBJECTIVES, GOALS, AND QUESTIONS

The issues surrounding responsibility, fairness, and equity in automated speech recognition (ASR) systems have been well documented in human-computer interaction (HCI), artificial intelligence (AI), machine learning (ML), natural language processing (NLP), and design research [2, 21, 23]. Notably, the usability of these systems heavily reliant upon language ML models trained on textual data and acoustic models trained on audio data. When the developed language models are trained on Standard[ized] American English (SAE) also known as Mainstream U.S. English (MUSE) (often denoted as the dominant variety of English in the United States), they perpetuate white, cis-normative, speech as preferable and socially acceptable [33]. This practice further echoes the practices and attitudes of racial majorities and MUSE groups, who often denigrate, reject, or discredit other minoritized varieties of spoken and even written English [30, 34]. Sociolinguist scholars suggest such patterns may contribute to further denigrating historical language varieties like African American Vernacular English (AAVE) (that are typically defined by race/ethnic groups) by categorizing them as ungrammatical and linguistically less meaningful than other forms of American speech, despite their important legacy in shaping American culture [29]. To this damaging effect, current approaches to technology development based on user-centered technology design currently face hurdles with recruitment and representation that may continue to re-entrench white, cis-normative perspectives within study designs [1, 3, 11]

Given these existing and potential challenges, I pose two primary research questions toward my outlined research agenda:

- What are the strengths and pitfalls of existing ASR system design processes, including the decisions NLP/ML technologists make that shape experiences of fairness and bias among underrepresented language variety users?
- How might researchers from academia and industry develop and employ collaborative-participatory approaches with African American community members — involving their voices and perspectives early and often in the product development process — to address many of the challenges African Americans (AAVE speakers) face when using voice technology?

4 RESEARCH APPROACH, METHODS, AND RATIONALE

To explore these questions, my research team and I will engage in a multi-phase research agenda that seek to prioritize principles of Design Justice in our pursuit of centering people who have been normally marginalized by design [8]. This work will seek to advance research on equitable and inclusive ASR system design through collaborative approaches across three phases: (1) understanding the norms and pitfalls in how ASR systems are designed, including the decisions that NLP/ML technologists make that contribute disparities among underrepresented language variety users; (2) partnering with local community organizations such as the King County Equity Now (KCEN) and the Africatown Community Landtrust (ACLT), whose missions center on advocating, liberating, and empowering Black communities, to engage in community-based participatory

research (CBPR) toward informing best practices within ASR system design; and (3) co-designing ASR and NLP prototypes with African American speakers of AAVE that inform the development of techniques for mitigating racial disparities in those systems [7, 13].

In Phase 1, our research team will engage NLP/ML technologists in interviews and focus groups to qualitatively understand the strengths and pitfalls of ASR system design and the decisions designers make that contribute to implications for disparities and bias among underrepresented language variety users. I expect to recruit 25-30 practitioners for interviews and invite roughly 20-25 for 4-5 focus groups by leveraging existing relationships with NLP/ML industry technologists (engineers, product managers, and researchers), while recruiting additional practitioners from a range of backgrounds based on product/system development experience, education, and demographics. I will conduct individual 60-minute virtual semi-structured interviews with these technologists, followed by comprehensive in-person focus groups (including both existing interviewees and additional contacts, as needed) to further contextualize normative practices that language system designers take that reveal implications for fairness related harms and bias in ASR systems. This phase will contribute empirical findings that illuminate design thinking processes.

In Phase 2, our research team will leverage our existing partnerships with two social justice and advocacy organizations in Seattle (King County), WA: 1) KCEN, a nonprofit focused on nurturing Black communities that own and control the resources and systems that impact Black lives; and 2) ACLT a non-profit providing equitable access to educational, technical and financial opportunities through technology development projects like the William Gross Center for African American Innovation. Drawing lessons from Phase 1, and building on our existing partnerships, I will use this phase to develop a shared set of values, goals, and activities for Phase 3. I will ensure mutual gains for all stakeholders by continuing our essential work to build rapport, share project motivations, and learn how human-centered design processes (within our department, university, and academic field) can support our partners' missions and aims [17]. I expect this support to take varied forms, including resources (monetary, equipment/technology, supplies, etc.), advocacy (visibility, ads, marketing), time (volunteering, serving alongside organizations, pro-bono design/research), ownership (study data, workshop artifacts, rights to disclose, reports). Extending our prior work to develop equitable and participatory approaches [10, 13, 31, 38], I will work with our partners to collectively define workshop conditions (locations, occurrences, dates/times, engagement norms) to ensure integrity and respect throughout our engagement with organizations.

In Phase 3, I will use findings from Phase 2 to iteratively develop a series of community-based participatory research (CBPR) "co-design" workshops aimed at addressing issues of fairness within ASR systems with AAVE speakers. The workshops will directly involve participants who are stakeholders of our partner organizations (KCEN, ACLT) who identify as African American speakers of AAVE. These participants will be positioned as co-design/research collaborators on our research team. The partner organizations will support recruitment by disseminating the study opportunity, recruiting participants, and ensuring sustainable engagement with our research team. During each workshop, I (Jay Cunningham) as

the research team lead and a designated peer leader nominated by the respective partner organization will co-facilitate a dialogue on the topic of mitigating bias and racial disparities of user experience in ASR and language systems for AAVE speakers. The outcomes of this phase will focus on producing a process for collecting high-quality, natural voice data for ML modeling that accurately represents the language of speakers. The co-facilitators will follow these critical conversations with an action-forward co-design session to explore and create language system prototypes that enable fair, equitable, and sustainable solutions that benefit all stakeholders. The workshops will take place over two academic quarters (roughly 20 weeks) within Seattle neighborhood locations most suitable for the organization's members (community centers, churches, schools, etc.) I aim to recruit and retain 10-15 members from each partner organization, appropriately compensating individual participants weekly for their collaborative engagement and by financially supporting our partner organizations. Data analysis will be conducted in collaboration with our partner organization to validate our findings. In turn, the findings from the community-based participatory research/design collaboration will be published in an academic journal and publicly circulated in a format well-suited for the partner organizations. Lastly, as collective co-research group, we'll best determine next steps for research and development that prompts sustainability in this critical effort to design language systems that are more responsible and inclusive.

5 RESULTS AND CONTRIBUTIONS TO DATE

Studies are currently underway or are in the process of being designed that aligns with this critical research agenda. My graduate research advisors Dr. Daniela Rosner, Dr. Julie Kientz, and I was awarded as recipients of the 2022 Google, LLC Award for Inclusive Research (AIR) for the agenda proposed in this research statement.

6 EXPECTED NEXT STEPS, DISCUSSION, CHALLENGES

Results from this three-phase study include: 1) informing practices of community-centered participation by understanding the experiences AAVE speakers; 2) developing collaborative research techniques with Black community organizations that help design more fair and equitable interaction experiences for AAVE users of language tech, 3) informing the public (Black AAVE speakers) of the adverse effects of racial disparities and to seek its elimination through increased technical knowledge collaborative partnerships with ML designers/researchers.

Over the past decade, efforts to understand the intersection of equity + technology have become more prioritized by academic, industry, non-profit, and government institutions [1]. Collectively, scholars, technologists, and designers alike seek to address design, development, and deployment of intelligent machines and their impacts on minority, marginalized, and vulnerable communities [10, 11, 19]. As an emerging responsible AI/ML UX researcher, I am holistically interested in understanding how complexities of technology design approaches, automated intelligent machines (AI/ML/NLP), and interactive systems impact, marginalize, and engage with underrepresented communities. **The intended contributions of my graduate research will include:** 1) Empirical

findings of the ways in which diverse language users (with an emphasis on Black American communities) are impacted by technical designs of language technologies, 2) Integrating and extending current human-centered design and participatory methodological frameworks as interventions toward more responsible and fair interactive language systems, and 3) Using novel equitable and inclusive approaches to design language technologies that directly influence technology practice and contribute exemplary technological artifacts. The collective broader impact advances technical design as tools of liberation and justice for underrepresented and marginalized language communities, not limited to and including speakers of ethnolect/sociolinguistic varieties (e.g., AAVE speakers), atypical speech (e.g., speech disabilities, speech impediments), and non-native English accents/dialects.

ACKNOWLEDGMENTS

This research is directly funded by the following awards:

National Science Foundation, Graduate Research Fellowship Program (Grant no: 1762114 | University of Washington)
Google, LLC., Award for Inclusion Research (AIR)

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